

Karnatak University Mathematics Teacher's Association (KUMTA)

B.Sc II Semester (RVSD NEP) Question Bank

Subject: MATHEMATICS Discipline Specific Core Course (DSC)

Course Title: Number Theory and Calculus

Course Code: C2 MAT 1 T1 (Theory)

Type of Course	Theory /Practical	Credits	Instruction hours per week	Total No. of Lecture	Duration of Sem End Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
DSC – 3	Theory	04	04	60 Hrs	03 Hrs	20	80	100

Syllabus

Unit – I **Number Theory:** **15 Hrs.**

Divisibility. Properties of divisibility. Division algorithm. GCD. Euclid's algorithm. Relatively prime numbers. Fundamental theorem of arithmetic. The number of positive divisors and sum of all the positive divisors of a number. The theory of congruence. Basic properties of congruence. Euler's theorem, Fermat's theorem. Wilson's theorem,

Unit – II **Integral Calculus:** **15 Hrs.**

Reduction Formulae, application of definite integrals to areas, volumes and surface of revolution. Length of plane curves.

Unit – III **Multivariate Calculus:** **15 Hrs.**

Functions of two or more variables, Partial derivatives of implicit and composite functions, Euler's theorem and its extension, Total differentials, Jacobians and standard properties and illustrative examples. Taylor's and Maclaurin's series for functions of two variables, Maxima-Minima of functions of two variables, Lagrange's method of undetermined multipliers.

Unit – IV **Vector Calculus:** **15 Hrs.**

Dot and cross product of vectors. Ordinary derivatives of vectors. Continuity and differentiability of a vector function. Derivatives of sum, dot product. Cross product and triple products of vectors. Differential of vectors. The vector differential operator del . The gradient, Divergence and curl. Solenoidal and irrational vectors.

Unit – I Number Theory:

I. Questions carrying Two Marks each

1. Define divisibility of two integers a and b

2. If a/b and b/c then prove that a/c
3. If a/b and a/c then prove that $a/b + c$
4. If a/b and a/c then prove that $a/b.c$
5. If a/b and x is any integer then (i) a/bx

6. Define division algorithm of two integers a and b

7. Define division g.c.d of two integers a and b.

8. State Fundamental theorem of arithmetic

9. Find the number of positive divisors of the following composite numbers.

(i) 90 (ii) 432 (iii) 824 (iv) 960 (v) 1800

10. Find the sum of all the positive divisors of the following composite numbers.

(i) 90 (ii) 432 (iii) 824 (iv) 960 (v) 1800

11. if $a \equiv b \pmod{m}$ & $b \equiv c \pmod{m}$ then $a \equiv c \pmod{m}$

12. if $a \equiv b \pmod{m}$ then x is any integer then prove that (any one)

i) $a + x \equiv b + x \pmod{m}$

ii) $a - x \equiv b - x \pmod{m}$

iii) $ax \equiv bx \pmod{m}$

13. if $a \equiv b \pmod{m}$ then $c \equiv d \pmod{m}$ then prove that (any one)

i) $a + c \equiv b + d \pmod{m}$

ii) $a - c \equiv b - d \pmod{m}$

14. if $ac \equiv bc \pmod{m}$ & $(c, m) = 1$ then prove that $a \equiv b \pmod{m}$

15. State Euler's theorem.

16. State Fermat's theorem.

17. State Wilson's theorem.

18. Find the number of integers less than 24 and relatively prime to 24

19. Find the number of integers less than 15 and relatively prime to 15

20. Find the number of integers less than 960 and relatively prime to 960.
21. Using Fermat's theorem Prove that $n^{16} - 1$ is divisible by 17 if $(n, 17) = 1$
22. Using Fermat's theorem find the remainder when 5^{10} is divided by 11.

II. Questions carrying Five Marks each

1. Prove that $9^n - 8^n - 1$ is divisible by 8.
2. Prove that $10^n + 3 \cdot 4^{n+2} + 5$ is divisible by 9.
3. Prove that $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24.
4. Find the g.c.d of 275 and 726. Write in the form $275x + 726y$
5. Find the g.c.d of 189 and 243. Write in the form $189m + 243n$ and show that m and n are not unique.
6. Find the g.c.d of 506 and 1155. Write in the form $506x + 1155y$
7. Find the g.c.d of 858 and 325. Write in the form $855m + 325n$
8. Find the g.c.d of 595 and 252. Write in the form $252m + 595n$
9. prove that a relation congruence is an equivalence relation .
10. if $a \equiv b \pmod{m}$ then x is any integer then
 - i) $a + x \equiv b + x \pmod{m}$
 - ii) $a - x \equiv b - x \pmod{m}$
 - iii) $ax \equiv bx \pmod{m}$
10. if $a \equiv b \pmod{m}$ then $c \equiv d \pmod{m}$ then
 - i) $a + c \equiv b + d \pmod{m}$
 - ii) $a - c \equiv b - d \pmod{m}$
 - iii) $ac \equiv bd \pmod{m}$

11. State and prove Fundamental theorem of arithmetic.

12. State and prove Euler's theorem.

13. State and prove Fermat's theorem.

14. Using Fermat's theorem prove that $2^{48} \equiv 1 \pmod{105}$
15. Using Fermat's theorem prove that $a^{18} - b^{18}$ is divisible by 133 where a and b are relatively prime to 133.

or

16. Prove that $a^{18} - b^{18} \equiv 0 \pmod{133}$

where a and b are relatively prime to 133.

17. Using Fermat's theorem prove that $x^{12} - y^{12}$ is divisible by 91

where x and y are relatively prime to 91.

18. State and prove Wilson's theorem.

20. Using Wilson's theorem prove that $p!$ and $(p-1)! - 1$ are relatively primes

if p is an odd prime number

21. Using Wilson's theorem, if p and q are primes then prove that

$$p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}.$$

22. Using Wilson's theorem prove that $(18+1)!$ is divisible by 437.

23. Using Wilson's theorem if p is prime number then prove that

$$2(p-3)! + 1 \equiv 0 \pmod{p}.$$

Unit – II Integral Calculus

I. Questions carrying Two Marks each

1. Evaluate $\int_0^{\frac{\pi}{2}} \sin^8 x \, dx$

2. Evaluate $\int_0^{\frac{\pi}{2}} \sin^{13} x \, dx$

3. Evaluate $\int_0^{\frac{\pi}{2}} \cos^{10} x \, dx$

4. Evaluate $\int_0^{\frac{\pi}{2}} \cos^{11} x \, dx$

5. Evaluate $\int_0^{\frac{\pi}{6}} \sin^6 3x \, dx$

6. Evaluate $\int_0^{\frac{\pi}{4}} \cos^6 2x \, dx$

7. Evaluate $\int_0^{\pi} \sin^7 \left(\frac{x}{2} \right) \, dx$

8. Evaluate $\int_0^{\infty} \frac{dx}{(1+x^2)^5}$

9. Evaluate $\int_0^{\infty} \frac{dx}{(1+x^2)^{\frac{7}{2}}}$

10. Evaluate $\int_0^{\frac{\pi}{2}} \sin^6 x \, dx \cos^4 x \, dx$

11. Evaluate $\int_0^{\frac{\pi}{2}} \sin^4 x \, dx \cos^7 x \, dx$

12. Evaluate $\int_0^{\pi} \sin^4 \left(\frac{\theta}{2} \right) \, dx \cos^2 \left(\frac{\theta}{2} \right) \, d\theta$

13. Find the reduction formula for $\int x^n e^{ax} \, dx$

14. Find the reduction formula for $\int a^x x^n \, dx$

15. Find the reduction formula for $\int (\log x)^n \, dx$

16 Find the length of an arc of the curve $y = \log \sec x$ from $x = 0$ to $x = \frac{\pi}{3}$

17. Find the length of an arc of the curve $y = \log \sec x$ from $x = 0$ to $x = \frac{\pi}{3}$

18. Find the area enclosed by the curve $y = 3x - x^3$ and the line $x = 0$ & $x = 3$

19. Find the area bounded by the parabola $y = 2x - x^2$ and x - axis
20. Find the area bounded by the x - axis and the curve $y = c \cosh\left(\frac{x}{c}\right)$ and the ordinates $x = 0, x = a$

II. Questions carrying Five Marks each

- Find the reduction formula for $\int \sin^n x \, dx$ and hence evaluate $\int \sin^3 x \, dx$
- Find the reduction formula for $\int \cos^n x \, dx$ and hence evaluate $\int \cos^4 x \, dx$
- Find the reduction formula for $\int \tan^n x \, dx$ and hence evaluate $\int \tan^3 x \, dx$
- Find the reduction formula for $\int x^m (\log x)^n \, dx$ and hence evaluate $\int x^3 (\log x)^2 \, dx$
- Obtain a reduction formula for $\int x^n e^{-x} \, dx$ and hence

show that $\int_0^{\infty} x^n e^{-x} \, dx = n!$

6. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ then

prove that (i) $I_n + I_{n-1} = \frac{1}{n+1}$ (ii) $n[I_{n-1} + I_{n+1}] = 1$

- Find the length of the arc of the parabola $y^2 = 4ax$ from the vertex to an extremity of the latus rectum (that is from $x = 0, x = 2a$).
- Find the length of the arc of the parabola $x^2 = 4ay$ from the vertex to an extremity of the latus rectum (that is from $x = 0, x = 2a$).
- Find the whole area of the circle $x^2 + y^2 = a^2$ by integration.
- Find the whole area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by integration.
- Find the whole area of the astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ by integration.

12. Find the area enclosed between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

13. show that area of the segment cut off from the parabolas $y^2 = 2x$ by the straight line $y = 4x - 1$ is $\frac{9}{32}$.

14. show that area bounded by the parabolas $x^2 = 8y$ and the line $x - 2y + 8 = 36$ sq units.

15. Find the volume of a sphere of radius a .

or

The circle $x^2 + y^2 = a^2$ is revolved about x - axis. Find the volume of the sphere so formed.

16. Find the volume of the solid generated by revolving the ellipse the

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ about the } x \text{ axis.}$$

17. Find the surface area of a sphere of radius a .

18. Find the surface area of the parabola $y^2 = 4ax$ revolving about the x - axis by the from the vertex to one end of the latus rectum.

19. Find the surface area of the solid generated by the revolution of the astroid

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$$

20. Find the surface area of the solid generated by the revolution the cycloid

$$x = a(\theta - \sin\theta), y = a(1 - \cos\theta).$$

Unit-3: Multivariate Calculus

15 Hrs.

Functions of two or more variables, Partial derivatives of implicit and composite functions, Euler's theorem and its extension, Total differentials, Jacobians and standard properties and illustrative examples. Taylor's and Maclaurin's series for functions of two variables, Maxima-Minima of functions of two variables, Lagrange's method of undetermined multipliers.

Two marks:

1. Discuss the continuity of the function $f(x, y) = \begin{cases} \frac{2xy}{\sqrt{x^2 + y^2}}, & (x, y) \neq 0 \\ 0, & (x, y) = 0 \end{cases}$.
2. Find $\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$ for $z = x^2 \sin(3x + y^2)$.
3. Verify $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ for $u = 3x^2 y - 3xy^2$.
4. Find $\frac{\partial^2 u}{\partial x \partial y}$ for $u = e^{x/y}$.
5. If $u = x^2(y - z) + y^2(z - x) + z^2(x - y)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
6. Define 'Homogeneous function' and give an example.
7. If $u = \log\left(\frac{x^2 + y^2}{x + y}\right)$ then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.
8. Find $\frac{du}{dt}$ for $u = x^2 y + xy^2$, $x = at^2$, $y = 2at$.
9. If $u = x + y$, $v = xy$ then find $\frac{\partial(u, v)}{\partial(x, y)}$.
10. Find the $\frac{\partial(x, y)}{\partial(r, \theta)}$ for $x = r \cos \theta$, $y = r \sin \theta$.
11. Find the $\frac{\partial(x, y)}{\partial(r, \theta)}$ for $x = u(1 + v)$, $y = v(1 + u)$.
12. Verify $J \times J' = 1$ for $x = u(1 - v)$, $y = uv$.
13. State Taylor's theorem for function of two variables.
14. State Maclaurin's theorem for function of two variables.
15. Define 'Maximum value' & 'Minimum value' for function of two variables.
16. Write the necessary conditions for extremum for function of two variables.
17. Write the sufficient condition for extremum for function of two variables.
18. Define 'Saddle point'.

Five marks:

1. If $u = x^2 \tan^{-1} \frac{y}{x} - y^2 \tan^{-1} \frac{x}{y}$ then find $\frac{\partial^2 u}{\partial x \partial y}$.
2. If $u = e^{ax + by} f(ax - by)$ then prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$.

3. Show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for $u = \log(x^2 + y^2)$.
4. If $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ then prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$.
5. If $x = r \cos \theta$, $y = r \sin \theta$ then prove that $\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r} \left\{ \left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 \right\}$.
6. If $u = f(r)$, $r = \sqrt{x^2 + y^2}$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$.
7. If $V = r^m$, $r = \sqrt{x^2 + y^2 + z^2}$ then prove that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m+1)r^{m-2}$.
8. State and prove Euler's theorem for homogeneous function of degree n in x & y .
9. If $u = f(x, y)$ be the homogeneous function of degree n in x & y then
- $$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u.$$
10. If $u = \tan^{-1} \left(\frac{x^2 + y^2}{x - y} \right)$ then show that
- $$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = (1 - 4\sin^2 u) \sin 2u$$
11. If $z = f(u, v)$ & $u = xy$, $v = \frac{y}{x}$ then show that $\frac{\partial z}{\partial x} + \left(\frac{y}{x} \right) \frac{\partial z}{\partial y} = 2y \frac{\partial z}{\partial u}$.
12. If $u = f(x - y, y - x, z - z)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.
13. If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_1 x_3}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$ then, show that $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = 4$.
14. If u, v are functions of x, y and x, y are again functions of r, θ then prove
- $$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)}$$
15. If $x = a(u + v)$, $y = b(u - v)$; $u = r^2 \cos 2\theta$, $v = r^2 \sin 2\theta$ then find $\frac{\partial(x, y)}{\partial(u, v)}$.
16. If J be the Jacobian of u, v w.r.t. x, y and J' be the Jacobian of x, y w.r.t. u, v then prove that $J \times J' = 1$.
17. If $u^3 + v^3 = x + y$; $u^2 + v^2 = x^3 + y^3$ then prove that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{(y^2 - x^2)}{2uv(u - v)}$.
18. Expand the function $f(x, y) = e^x \sin y$ by Maclaurin's theorem up to third degree terms.
19. Expand the function $f(x, y) = \sin x \cos y$ by Maclaurin's theorem up to third degree terms.
20. Discuss the extreme values for the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.
21. Find the extreme values for the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.
22. Find the extremum value for $x^2 + y^2 + z^2$ subject to $xyz = a^3$

Unit-4: Vector Calculus

15 Hrs.

Dot and cross product of vectors. Ordinary derivatives of vectors. Continuity and differentiability of a vector function. Derivatives of sum, dot product. Cross product and triple products of vectors. Differential of vectors. The vector differential operator ∇ . The gradient, Divergence and curl. Solenoidal and irrotational vectors.

Two marks:

1. A particle moving along a curve $x = 3t^2$, $y = t^2 - 2t$, $z = t^3$ then determine the velocity and acceleration at any time t & their magnitudes at $t = 1$.
2. Evaluate $\frac{d}{dt}(\vec{A} \cdot \vec{B})$ & $\frac{d}{dt}(\vec{A} \times \vec{B})$ for $\vec{A} = 5t^2\hat{i} + t\hat{j} + t^3\hat{k}$ & $\vec{B} = \sin t\hat{i} - \cos t\hat{j}$.
3. If $\vec{A} = e^{-t}\hat{i} + \log(1 + t^2)\hat{j} - \tan t\hat{k}$ then find $\left| \frac{d^2\vec{A}}{dt^2} \right|$ at $t = 0$.
4. If $\vec{A} = (1 - \cos t)\hat{i} + (t - \sin t)\hat{j}$ then find $\vec{A} \cdot \frac{d\vec{A}}{dt}$.
5. If the vector function $\vec{f}(t)$ to be constant if and only if $\frac{d\vec{f}}{dt} = \vec{0}$.
6. Verify $\frac{\partial^2 \vec{f}}{\partial x \partial y} = \frac{\partial^2 \vec{f}}{\partial y \partial x}$ for $\vec{f} = (2x^2y - x^4)\hat{i} + (e^{xy} - y\sin x)\hat{j} + x^2\cos y\hat{k}$.
7. If f & g be the differentiable scalar functions of the variables x, y, z then prove that $\nabla(fg) = f(\nabla g) + g(\nabla f)$.
8. Find the gradient of the scalar function $\phi = xyz^2$ at $(2, 1, 1)$.
9. Find the directional derivative of $\phi = x^2 + y^2 + z^2$ at the point $(1, 2, 3)$ in the direction of $3\hat{i} + 4\hat{j}$.
10. If $\vec{A}$ & $\vec{B}$ be the differentiable vector functions of the variables x, y, z then prove that $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$.
11. Find the divergence of the function $\vec{f} = x^2z\hat{i} - 2y^3z^2\hat{j} + xy^2z\hat{k}$ at $(1, -1, 1)$.
12. If $\phi = 3x^2 - yz$ then find $\nabla \cdot (\nabla \phi)$.
13. Show that $\nabla \phi \cdot d\vec{r} = d\phi$.
14. Prove that $\text{div}(\text{grad } \phi) = \nabla^2 \phi$.
15. If $\vec{A}$ & $\vec{B}$ be the differentiable vector functions of the variables x, y, z then prove that $\nabla \times (\vec{A} - \vec{B}) = \nabla \times \vec{A} - \nabla \times \vec{B}$.
16. If $\vec{A} = x^2y\hat{i} + y^2z\hat{j} + z^2x\hat{k}$ then find $\nabla \times \vec{A}$ at $(1, 2, 3)$.
17. Define Solenoidal & irrotational of vector point function.
18. Prove that the vector function $\vec{f} = (x + 3)\hat{i} + (y - 2z)\hat{j} + (x - 2z)\hat{k}$ is Solenoidal

Five marks:

1. If $\vec{A}$ & $\vec{B}$ be the vector functions of the scalar variable t then prove that

$$\frac{d}{dt}(\vec{A} + \vec{B}) = \frac{d\vec{A}}{dt} + \frac{d\vec{B}}{dt}.$$

2. If ϕ be the scalar function of t & $\vec{A}$ be the vector function of the scalar variable t then

$$\frac{d}{dt}(\phi \bar{A}) = \phi \frac{d\bar{A}}{dt} + \bar{A} \frac{d\phi}{dt}.$$

3. If $\bar{A}$ & $\bar{B}$ be the vector functions of the scalar variable t then

$$\frac{d}{dt}(\bar{A} \cdot \bar{B}) = \bar{A} \cdot \frac{d\bar{B}}{dt} + \bar{B} \cdot \frac{d\bar{A}}{dt}.$$

4. If $\bar{A}$ & $\bar{B}$ be the vector functions of the scalar variable t then

$$\frac{d}{dt}(\bar{A} \times \bar{B}) = \bar{A} \times \frac{d\bar{B}}{dt} + \frac{d\bar{A}}{dt} \times \bar{B}.$$

5. Find $\frac{d}{d\theta}\{\bar{a} \times (\bar{b} \times \bar{c})\}$ at $\theta = 0$ for

$$\begin{aligned}\bar{a} &= \sin\theta \hat{i} + \cos\theta \hat{j} + \theta \hat{k}, \quad \bar{b} = \cos\theta \hat{i} - \sin\theta \hat{j} - 3\hat{k}, \\ \bar{c} &= 2\hat{i} + 3\hat{j} - \hat{k}.\end{aligned}$$

6. If $\bar{r} = \cos nt \hat{i} + \sin nt \hat{j}$, n is a constant then prove that $\bar{r} \times \frac{d\bar{r}}{dt} = n\hat{k}$.

7. If $\bar{r} = \cos\omega t \bar{a} + \sin\omega t \bar{b}$, where $\bar{a}$ & $\bar{b}$ are constant vectors and ω is a scalar constant then prove that

$$\frac{d^2\bar{r}}{dt^2} + \omega^2 \bar{r} = \bar{0} \quad \& \quad \bar{r} \times \frac{d\bar{r}}{dt} = \omega(\bar{a} \times \bar{b}).$$

8. If $\bar{r} = a \cos t \hat{i} + a \sin t \hat{j} + t \hat{k}$ then find $\left[\bar{r} \cdot \frac{d\bar{r}}{dt} \cdot \frac{d^2\bar{r}}{dt^2} \right]$ at $t = \pi/4$.

9. If the vector function $\bar{f}(t)$ to have constant magnitude if and only if $\bar{f} \cdot \frac{d\bar{f}}{dt} = 0$.

10. The vector function $\bar{f}(t)$ to have constant direction if and only if $\bar{f} \times \frac{d\bar{f}}{dt} = \bar{0}$.

11. Find the gradient of the function $\phi = r^n$, $r = \sqrt{x^2 + y^2 + z^2}$.

12. If $\bar{A}$ be the differentiable vector function of the variables x, y, z then prove that $\nabla \times (\phi \bar{A}) = \phi(\nabla \times \bar{A}) + (\nabla \phi) \times \bar{A}$, ϕ is a differentiable scalar function of the variables x, y, z .

13. If $\bar{A}$ be the differentiable vector function of the variables x, y, z then prove that

$$\nabla \times (\nabla \times \bar{A}) = -\nabla^2 \bar{A} + \nabla(\nabla \cdot \bar{A})$$

14. If $\bar{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then find $\nabla \cdot \bar{r}$ & $\nabla \times \bar{r}$.

15. If $\bar{f} = (x^2 + yz)\hat{i} + (y^2 + xz)\hat{j} + (z^2 + xy)\hat{k}$ then find

$$\text{div } \bar{f} \quad \& \quad \text{curl } \bar{f}.$$

16. If $\bar{f} = (x + y + 1)\hat{i} + \hat{j} - (x + y)\hat{k}$ then prove that $\bar{f} \cdot \text{curl } \bar{f} = 0$.

17. Show that $\text{div}(\text{grad } \phi) = 0$, $\phi = \frac{1}{r}$.

18. Show that $\text{curl}(\text{grad } \phi) = 0$, $\phi = \frac{1}{r}$.

19. Show that $\text{div}(\text{curl } \bar{A}) = 0$, $\bar{A}$ is any vector function.

20. Find $\text{curl } \bar{P}$, where $\bar{P} = e^{xyz}(\hat{i} + \hat{j} + \hat{k})$.

21. If $\vec{V} = x^2 y \hat{i} - 2xz \hat{j} + 2yz \hat{k}$ then $\text{curl}(\text{curl} \vec{V})$.
22. If $\vec{a}$ is a constant vector. Prove that $\text{div}(\vec{a} \times \{\vec{r} \times \vec{a}\}) = 2|\vec{a}|^2$.
23. If $\vec{a}$ & $\vec{b}$ are constant vectors then prove that $\text{curl}(\{\vec{r} \times \vec{a}\} \times \vec{b}) = \vec{b} \times \vec{a}$.
24. Find the constants a , b , c so that the vector $\vec{V}$ is irrotational
- $$\vec{V} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$$